

Navigating Excel, Formatting & Cell References

INSTRUCTOR SOLUTION — DO NOT DISTRIBUTE TO STUDENTS

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SOLUTION KEY

Sample answers below. Accept any student response that correctly identifies the key concept. Grading guidance: each problem is worth equal weight unless otherwise noted on the assignment rubric.

Problem 1

You open a new Excel workbook. Identify the two screen elements that together tell you (a) which cell is currently selected and (b) whether that cell contains a typed value or a formula. Name both elements and briefly explain what each one shows.

Answer: The Name Box (left of the formula bar) shows the address of the selected cell, e.g., C3. The Formula Bar shows the raw content of that cell — if the cell contains a formula, the formula bar displays the formula (e.g., =B3*\$B\$1), while the cell itself displays only the result.

Problem 2

Tidal Goods Co. has a product pricing spreadsheet with 300 rows of data. List the keyboard shortcut you would use to (a) jump to the last used cell in the worksheet and (b) return to cell A1. For each shortcut, describe what it does in one sentence.

Answer: (a) Ctrl+End — jumps to the last used cell in the worksheet (the bottom-right corner of the data). (b) Ctrl+Home — returns to cell A1, the top-left corner of the worksheet.

Problem 3

A Tidal Goods Co. analyst types a 15% seasonal discount rate as 0.15 into cell D1 and leaves the cell unformatted. Explain what problem this creates when someone else reads the spreadsheet. What formatting step should the analyst take, and how do they apply it in Excel?

Answer: Without formatting, a reader cannot determine whether 0.15 represents 15%, \$0.15, or some other unitless value. The analyst should apply Percentage format: select cell D1, go to the Home tab, and choose Percentage from the Number group. The cell will then display 15%.

Problem 4

The formula $=B4*(1+B1)$ is in cell C4 of a Tidal Goods pricing worksheet. B1 contains the markup rate (40%) and B4 contains a product cost (\$22.50). When this formula is copied to cell C5, what does Excel place in C5? Show the formula and explain why each reference changed or stayed the same.

Answer: Excel places $=B5*(1+B2)$ in C5. Because both references are relative, each shifts one row down when the formula is copied. B4 becomes B5 (the next product's cost — correct). B1 becomes B2 (no longer the markup rate cell — this is a mistake, which is why B1 should be written as $\$B\1).

Problem 5

Rewrite the formula from Problem 4 so that the markup rate always references cell B1, regardless of where the formula is copied. Explain which reference type you used and why.

Answer: The corrected formula is $=B5*(1+\$B\$1)$. The cost reference (B5) remains relative so it updates correctly for each product row. The markup rate reference ($\$B\1) uses absolute reference — dollar signs lock both the column (B) and the row (1), so no matter where the formula is copied, it always reads the 40% rate from B1.

Problem 6

You are entering a formula and type $=C3$ as part of it. Without lifting your hands from the keyboard, how do you change this to an absolute reference? Describe the key sequence and the result after the first press.

Answer: With the cursor immediately after C3 in the formula bar (or while still typing), press F4 once. Excel changes $=C3$ to $=\$C\3 , locking both the column (C) and the row (3). Pressing F4 again cycles to $=C\$3$ (row locked only), then to $=\$C3$ (column locked only), then back to $=C3$ (fully relative).